Md Safaiat Hossain

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EDUCATION

Ph.D. in data Science and Engineering — Knoxville, Tennessee
The University of Tennessee, Advisor: Dr. Xiao Wang — August 2024 - Current

Bachelor of Science in Electrical and Electronic Engineering — Natore, Bangladesh
Bangladesh Army university of Engineering and Technology — March 2016 - March 2020

- Thesis: Alphabet Recognition by Analyzing Brain Signals & Classifying by Use of Machine Learning. (Advisor: Md. Mustafizur Rahman)

PUBLICATIONS

Book Chapters:

[1] Wang, X., Rajagopal, J., Hossain, M. S., Chen, P., Wahib, M., Zhang, E., & Reid, E. J. (2026). *Efficient Computing for Medical Image Acquisition and Reconstruction*. Book chapter for Medical Image Vision Handbook. Available as arXiv:2607.13204. https://doi.org/10.48550/arXiv.2607.13204

[2] Karim, M. B., Hossain, M. S., Roky, R. R., & Srizon, A. Y. (2025). BHSGR-Net: A Light-Weight Convolutional Neural Architecture for Recognition of Bengali Hand Sign Gestures. In *Data Driven Applications for Industry 4.0 and Beyond* (1st ed.). Chapman & Hall/CRC, Taylor & Francis.

Conference papers:

[1] M. S. Hossain, M. T. Hassan and M. S. Hossain, "Biologically Inspired Network Design to Train Only Through Forward Pass," *2024 2nd International Conference on Information and Communication Technology (ICICT)*, Dhaka, Bangladesh, 2024, pp. 130-134.

[2] M. S. Hossain, M. S. Hossain and M. B. Karim, "Training StyleGAN2 on BWW-Texture Dataset: Generating Textures Through Transfer Learning," *2023 International Conference on Information and Communication Technology for Sustainable Development (ICICT4SD)*, Dhaka, Bangladesh, 2023, pp. 341-345.

[3] R. B. Angan, M. S. Hossain, S. S. Nobel and M. M. Hasan, "Health Monitoring of Old Buildings in Bangladesh: Detection of Cracks and Dampness Using Image Processing," *2023 International Conference on Next-Generation Computing, IoT and Machine Learning (NCIM)*, Gazipur, Bangladesh, 2023, pp. 1-5.

[4] M. S. Hossain, R. B. Angan and M. M. Hasan, "Pothole Detection and Estimation of Repair Cost in Bangladeshi Street: AI-based Multiple Case Analysis," *2023 International Conference on Electrical, Computer and Communication Engineering (ECCE)*, Chittagong, Bangladesh, 2023, pp. 1-6.

[5] M. S. Hasan, M. S. Hossain and M. R. Hayder, "Converting Municipal Solid Waste into Electrical Energy: A Renewable Solution in Bangladesh," *2023 International Conference on Electrical, Computer and Communication Engineering (ECCE)*, Chittagong, Bangladesh, 2023, pp. 1-5.

[6] M. S. Hossain, T. Hasan, M. M. Hasan, M. M. Rahman and M. M. Sabiha, "English Character recognition using EEG-based Visual stimulations: A Machine Learning Analysis," *2022 International Conference on Innovations in Science, Engineering and Technology (ICISET)*, Chittagong, Bangladesh, 2022, pp. 373-378.

Poster Presentation

[1] R. B. Angan, M. S. Hossain and S. Nobel, "Damage detection of old structure with deep learning: An onsite survey," in *4th International Conference on Sustainable Technologies for Industry 4.0 (STI 2022)*, Dhaka, 2022.

Experience

Oak Ridge National Laboratory Oak Ridge, Tennessee
Graduate Research Assistant (Supervisor: Dr. Xiao Wang) August 2024 - Current

- Developing physics-informed model-based iterative reconstruction (MBIR) algorithms for SPECT, PET, and CT imaging.
- Researching AI-enhanced tomographic reconstruction using optimization, uncertainty quantification, and generative priors.

Rajshahi University of Engineering and Technology Rajshahi, Bangladesh
Research Assistant (Supervisor: Azmain Yakin Srizon) December 2022 - December 2023

- Developed Bangla Hand Sign Gestures Dataset for muted people
- Bangla Hand Sign recognition system

Dhaka, Bangladesh
November 2023-July 2024

Bangladesh Polytechnic Institute Rajshahi, Bangladesh
Instructor May 2022-September 2023

- Experienced in teaching electrical and electronics courses with a strong emphasis on practical applications.
- Proficient in designing course materials, including assignments, homework, and exams.
- Experienced in evaluating student performance through assessments and providing constructive feedback.
- Courses Taught: Transmission and Distribution of Electrical Power I & II, Microcontroller and PLC, Microcontroller and Embedded Systems, Renewable Energy

Bangladesh Betar Rajshahi, Bangladesh
Radio Broadcasting and Transmitting System Intern February 2019

- Assisted in the setup, maintenance, and troubleshooting of radio transmission systems
- Monitored signal quality and ensured smooth broadcasting operations
- Conducted testing and calibration of broadcasting equipment for optimal performance

Skills

- Programming Language: Python, C, C++
- Library: NumPy, Pandas, Torch, scikit-learn,
- Modeling and Simulation: Matlab
- Software & Tools: Git, Minitab, Anaconda